



DATA MOSAIC

Data Engineering

Abstract

Crypto Currency data mosaic, sourcing data from well known sources. Providing insights and charts to showcase trends

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Contribution Details

Project Overview:

- Extracted cryptocurrency data from Reddit, Google Trends, Kaggle, and Yahoo.
- Tasks included data extraction, cleaning, exploratory analysis, visualization, coding, and report writing.

Division of Responsibilities:

- **Data Extraction:**
 - *Ahsan Khan:* Reddit & Kaggle
 - *Faizan Ullah:* Google Trends & Yahoo
- **Data Cleaning & Integration:**
 - Collaboratively cleaned and merged all datasets to maintain consistency.
- **Exploratory Analysis & Visualization:**
 - Jointly analysed data to uncover key trends and insights.
 - Worked together to design and refine visualizations.
- **Coding & Peer Review:**
 - Both members developed initial code for all four data pipelines.
 - Exchanged code for peer review, formatting and to ensure quality and mutual understanding.
- **Report Writing:**
 - Shared the effort equally in drafting and finalizing the project report.

Conclusion:

- Our balanced approach and collaborative efforts ensured equal contribution across all aspects of the project.

Overview:

We selected cryptocurrency as our topic because it is currently a hot discussion area in both technology and finance. Relevant data is often available, making it accessible for analysis and modelling. Machine learning models can be used to predict trends and help in investment decision-making.

Expected Data: We anticipated to find metrics like:

- Historical price data
 - Daily prices, trade volumes, market capitalization).
- User interest or sentiment:
 - Reddit posts discussing particular coins, public opinion).
- Search trends:
 - Indicating popularity or public curiosity over time via google trends

Data Collection Process:

We collected data from **three main sources**:

1. Reddit

- **API:** Created an API application on Reddit.
- **Library:** Used the praw Python library to handle data extraction.
- **Sample Size:** Fetched *100 posts*, adhering to the required sample size and respecting rate limits.

2. Public Datasets (Kaggle)

- **Datasets:**
 1. [*Dataset 1*](#)
 2. [*Dataset 2*](#)
- **Access:** Downloaded manually from Kaggle for offline analysis with Pandas, furthermore, we wrote a script to automate processing of dataset 2 source from kaggle.
- **Challenges:** *[Insert any specific obstacles you faced (e.g., data cleaning, large file sizes, etc.)]*

3. Yahoo API

- **Tickers:** BTC-USD, ETH-USD, DOGE-USD, SOL-USD, BNB-USD.
- **Timeframe:** Gathered *2 years of historical data* (daily prices, volumes, etc.).
- **Process:** Used Python's data fetching libraries (e.g., yfinance) to download and then analyze with Pandas.

4. Google Trends

- **Library:** pytrends was used to fetch interest over time data for keywords like "Bitcoin" and "Crypto."
- **Timeframes:** Divided into monthly segments from *[October 2024 to January 2025]* due to rate limit constraints.

Challenges Encountered:

- *API rate limits* on Reddit, leading us to moderate our requests.
- *API rate limits* on Google with 429 error, leading us to pivot to other data source, like yahoo.
 - **HTTP 429 Rate Limit Error:** Google often blocked large requests, so we spaced out requests with a 3-minute sleep (`time.sleep(180)`).
 - **Partial Data:** Couldn't collect all months in one go; had to break into smaller intervals.
 - Needed to **clean/merge** monthly CSV files afterward.
- *Incomplete or inconsistent data* in certain Kaggle files.
- *Terms of Service (TOS)* constraints: We had to ensure we respect usage limits and privacy guidelines.

Initial Observations:

Reddit:

- **File:** crypto_reddit_data.csv
- **Shape:** 100 rows × 6 columns

Observations:

- There are 40 non-null entries under "Post Text," indicating many posts might be link-only or have no body text.
- Dates are stored as object. We may want to convert them to a proper datetime type for time-based analysis.

- No missing values in Title, Author, or Subreddit, but do note that “Post Text” has only 40 non-null values.

Snippet:

```
files = glob.glob("*reddit*.csv")
print(files)
df = pd.read_csv(files[0])
print(df.info())
```

[19] ✓ 0.0s

```
... ['crypto_reddit_data.csv']
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Title       100 non-null    object
1   Post Text   40 non-null     object
2   Author      100 non-null    object
3   Date        100 non-null    object
4   Upvotes     100 non-null    int64
5   Subreddit   100 non-null    object
dtypes: int64(1), object(5)
memory usage: 4.8+ KB
None
```

Kaggle:

- **Files:** Multiple CSVs (e.g., "crypto_coin1.csv", "crypto_coin2.csv", etc.), concatenated into merged_df.
- **Shape:** 37,082 rows × 10 columns (after merging).

Observations:

- All columns have 37,082 non-null entries, indicating no immediate NaNs.
- Dates are stored as object; converting to a proper datetime type is recommended.
- Typical float columns for numeric metrics; potential for deeper EDA, such as computing average close prices, total marketcap by date, etc.

```
import glob
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Get a list of all files that start with "crypto" and end with ".csv"
files = glob.glob("coin*.csv")
files

df_list = [pd.read_csv(file) for file in files] # Read each file into a DataFrame
merged_df = pd.concat(df_list, ignore_index=True) # Merge them into one
# Display the first few rows
merged_df.info()
```

[37] ✓ 0.9s

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 37082 entries, 0 to 37081
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   SNo          37082 non-null  int64
1   Name         37082 non-null  object
2   Symbol       37082 non-null  object
3   Date         37082 non-null  object
4   High         37082 non-null  float64
5   Low          37082 non-null  float64
6   Open         37082 non-null  float64
7   Close        37082 non-null  float64
8   Volume       37082 non-null  float64
9   Marketcap    37082 non-null  float64
dtypes: float64(6), int64(1), object(3)
memory usage: 2.8+ MB
```

Google Trends:

- **Files:** For each month in *November 2024 through January 2025*, we have CSV files named like `crypto_trends_november_2024.csv`, `crypto_trends_december_2024.csv`, `crypto_trends_january_2025.csv`, etc.
- **Shape:** Each file typically has ~ 45 rows $\times$ ~ 4 columns (depending on the requested timeframe). When merged with the Reddit file (in the screenshot), the combined DataFrame shows 145 entries $\times$ 10 columns.

Observations:

- Some columns overlap with the Reddit dataset (Title, Post Text, etc.) even though they are not relevant to Google Trends. This suggests we merged all CSVs into one DataFrame. In practice, we might prefer separate DataFrames or carefully join on date/time or a common key.
- The `isPartial` column is often True or False; partial data might skew averages if we're doing monthly comparisons.
- Rate limit (HTTP 429) caused partial coverage: we only successfully retrieved half-month data for each month.

```
import plotly.graph_objects as go
files = glob.glob("crypto*.csv")
print(files)

df_list = [pd.read_csv(file) for file in files] # Read each file into a DataFrame
merged_df = pd.concat(df_list, ignore_index=True) # Merge them into one
# Display the first few rows
print(merged_df.info())
df = merged_df

[38] ✓ 0.0s

... ['crypto_reddit_data.csv', 'crypto_trends_december_2024.csv', 'crypto_trends_january_2025.csv', 'crypto_trends_november_2024.csv']
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 145 entries, 0 to 144
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Title                  100 non-null   object
1   Post Text              40 non-null    object
2   Author                 100 non-null   object
3   Date                   100 non-null   object
4   Upvotes                100 non-null   float64
5   Subreddit              100 non-null   object
6   date/time              45 non-null    object
7   Bitcoin_interest_score 45 non-null    float64
8   Crypto_interest_score  45 non-null    float64
9   isPartial              45 non-null    object
dtypes: float64(3), object(7)
memory usage: 11.5+ KB
None
```

AI Product preposition:

Our goal is to develop an AI/ML model that can:

- **Forecast Cryptocurrency Prices:** Integrate historical price data with social media sentiment (Reddit) and global interest (Google Trends).
- **Provide Investment Insights:** A dashboard or platform that highlights correlations as if high Reddit activity or Google Trends spikes correlate with price swings.

Terms of Service & Privacy Considerations.

- **Reddit:**
 - Must not collect or store personally identifiable information.
 - Rate limits for API calls must be respected.
- **Google:**
 - Rate-limiting encountered (429 error) required us to split requests.
 - Must not redistribute raw user-generated or user-specific data in a way that violates Google's TOS.
- **GDPR Compliance:**
 - Anonymize or exclude any personal data (especially from Reddit).
 - Retain only aggregated or publicly available information.

Multiple Data Sources: Benefits & Conflicts

Aspect	Helps Data Quality	Hinders Data Quality
Coverage & Completeness	Broader perspective, reduced data gaps.	Possible missing/incomplete data or differing data periods.
Validation & Reliability	Can cross-check info (e.g., Reddit vs. Google Trends).	Conflicting insights if sources don't agree or have timing mismatches.
Richness & Diversity	Combines structured (Kaggle) & unstructured (Reddit).	Requires heavy preprocessing, more complex pipelines.
Different Data Formats	Varied perspectives (API, CSV, JSON).	Data must be standardized (dates, naming conventions, etc.).
Timeframe Mismatch	Multiple intervals (monthly, daily) provide deeper view.	Misaligned timestamps can distort correlation or trend analysis.
Rate Limit Constraints	Slows data collection, ensures better compliance.	Potential for partial or incomplete data if time-limited.

Storing and Combining the Data

Relational Database:

- Separate tables (e.g., reddit_data, google_trends_data, kaggle_price_history) linked via date or cryptocurrency keyword.
- Joins: Data can be combined with queries for time-based or coin-based analyses.

Object-Based Storage – S3:

- CSV files in a well-structured hierarchy (dataset/raw/, dataset/processed/).
- Easy to share and version control, but may be less efficient for complex queries.

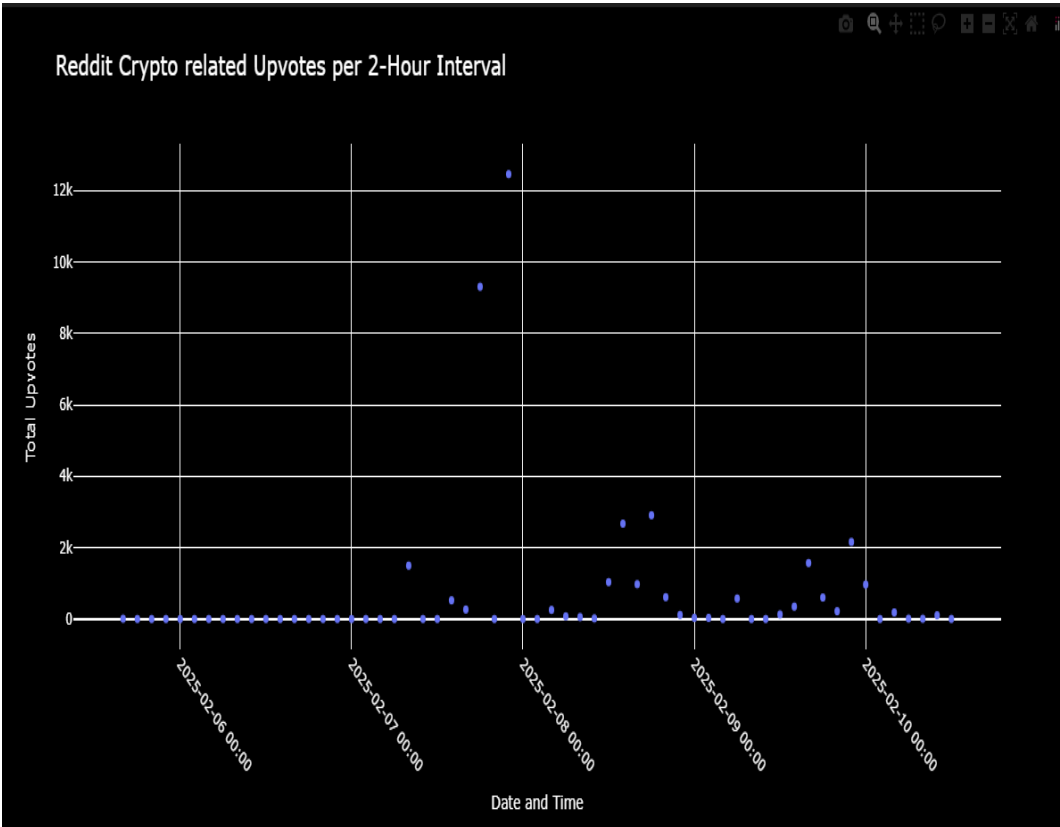
Potential NoSQL:

- Might store Reddit text data in a document store (e.g., MongoDB) for flexible indexing and text search.

Visualizations (Optional).

Reddit:

A scatter plot of total Reddit crypto-related upvotes aggregated every two hours, highlighting periods of intense user engagement



Kaggle Datasets:

A table displaying mean, minimum, and maximum daily price changes for various cryptocurrencies

Name	Price Change Analysis		
	mean	min	max
Aave	1.020890396	-210.8623443	100.1452065
Binance Coin	0.222076266	-169.8372467	136.6281765
Bitcoin	11.91027347	-7563.999292	7309.636429
Cardano	0.001026032	-0.52406206	0.36521863
Chainlink	0.014102063	-15.935869	6.48843066
Cosmos	0.012294858	-8.91786611	3.79771406
Crypto.com Coin	0.000115191	-0.05798204	0.07964223
Dogecoin	9.35E-05	-0.14331189	0.18428271
EOS	-0.00037543	-4.2124338	5.09629752
Ethereum	1.03079225	-921.9779494	543.6550803
IOTA	-9.14E-05	-1.628550053	1.122340202
Litecoin	0.026750067	-105.0854048	103.6949821
Monero	0.065575362	-141.5504155	93.48400679
NEM	-4.06E-05	-0.35066998	0.642096977
Polkadot	0.026572354	-15.46167641	7.97638834
Solana	0.081359529	-20.96207346	9.69970153
Stellar	9.61E-05	-0.19644108	0.333507001
TRON	3.16E-05	-0.053201988	0.113053903
Tether	3.98E-05	-0.053789673	0.059059978
USD Coin	-5.62E-07	-0.031542048	0.042768668
Uniswap	0.062477843	-11.64325616	9.06774094
Wrapped Bitcoin	34.03847443	-7586.297132	7534.523221
XRP	0.00024935	-0.907360077	0.779030136

A table summarizing mean, minimum, and maximum volatility measures for each cryptocurrency.

Volatility Analysis			
Name	mean	min	max
Aave	33.67380442	2.21790142	294.5923428
Binance Coin	5.598287263	0.003849797	232.262867
Bitcoin	440.5667001	1.02799878	12864.61957
Cardano	0.030096908	0.000613	1.00006179
Chainlink	0.712389105	0.007707	22.23466097
Cosmos	0.781104792	0.040086677	13.62042325
Crypto.com Co	0.006908963	0.000196617	0.13760279
Dogecoin	0.002570833	2.26E-06	0.4664675
EOS	0.454053019	0.017242968	5.62694304
Ethereum	32.66587936	0.018745005	1485.475556
IOTA	0.091329688	0.002835692	2.308480263
Litecoin	4.690408682	0.009279966	148.6751564
Monero	6.876140967	0.010319978	191.7178966
NEM	0.015864111	1.81E-06	0.671925008
Polkadot	2.222558327	0.09336397	22.70520559
Solana	1.733558974	0.021136248	28.0959122
Stellar	0.011199505	2.34E-05	0.47869244
TRON	0.003751284	0.00011634	0.123650998
Tether	0.01110563	0	0.209370017
USD Coin	0.011694288	0.00036165	0.131202521
Uniswap	2.156862652	0.12514779	18.81876966
Wrapped Bitco	1289.050838	91.25845605	32556.4284
XRP	0.025767506	0	1.298049927

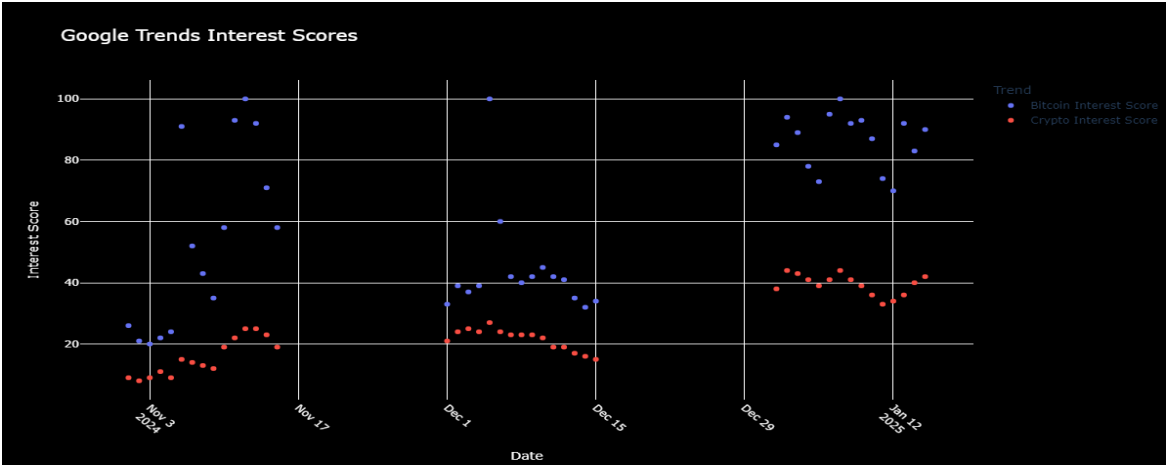
Yahoo (“Close” statistics):

A snapshot showing each ticker’s mean, minimum, and maximum prices over the queried timeframe.

A	B	C	D
Ticker	Mean	Min	Max
BNB-USD	427.5287015	205.2294159	750.272644
BTC-USD	52107.56612	20187.24414	106146.2656
DOGE-USD	0.133603762	0.057897002	0.467296988
ETH-USD	2518.743167	1429.158081	4066.445068
SOL-USD	104.0352153	14.48657227	261.869751

Google Trends:

Scatter plots indicating ‘Bitcoin’ and ‘Crypto’ interest scores over time, as fetched via pytrends



Yahoo:

A line chart comparing the 4-month average prices for multiple crypto tickers (BNB, BTC, DOGE, ETH, SOL)

